Shwetank Agrawal

Ph.D. – IIT Roorkee || M.Tech – IIT BHU || B.Tech – NIT Rourkela || Research Intern – GE Vernova ARC || GET – JSW

DGITIAL PORTFOLIO

SCAN TO CONNECT WITH ME

Contact

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Area of Interest

Mathematical modeling & Stability analysis, Microgrid Operation & Control, Automatic Generation and Control, Design and control of Power Electronics Converters, Integration of Renewable energy sources like Solar and wind with the grid, Implementation of grid codes, grid forming, feeding, and supporting control algorithms for inverters, centralized, distributed control implementation in Microgrid. Validating control concept through SIL/HIL/CHIL.

Core Competencies

- ✓ **Programming Languages:** MATLAB Scripting, C/C++, Python, Ladder logic, Verilog-Basic.
- ✓ **Software Packages:** MATLAB/ Simulink, RT-LAB, PSCAD, PSSE, Code Composer Studio, STM32CubeIDE, Arduino IDE, LTSpice, KiCAD, LabView, PLECS, Git.
- ✓ **Controllers:** TI-C2000, STM32-F446RE, Opal-RT, dSPACE, Arduino, S7-400 PLC, NI-DAQ.
- ✓ **Communication:** UART, SPI, Modbus Protocol, IEC 61850 Standard.

Educational Record

S. No.	Degree	Institution/Board	Year	Percentage/CGPA
1	Ph.D. (Power Electronics converter Design & Control in Power System)	IIT Roorkee, India	2019 – 2026	8.857/10
2	M.Tech (Power System)	IIT(BHU), Varanasi, India	2014 – 2016	8.41/10
3	B.Tech (Electrical Engineering)	NIT Rourkela, India	2009 – 2013	8.66/10
4	12^{th} Class	U.P. Board	2007-2008	68.00%
5	10^{th} Class	U.P. Board	2005-2006	76.66%

Professional Experience (1 year, 3 Months)

✓ **Research Intern – Electrical – Power Electronics and Power System**
GE Vernova Advance Research Center Bengaluru (Mar 2025 – Nov 2025)

- PowerUp Project – To uprate the Wind Turbine, define operating limits through D-Curve and temperature margins through SS Tool with DTMA.
- Running SIL setup for the Wind Turbine System.
- Improve diagnostics for CCU Project: Communication between the FPGA and the GE Marksat controller.
- Grid Seismograph Project: SSO detection through EKF.

✓ **Graduate Engineer Trainee (GET)**
JSW Steel Limited, Dolvi, Mumbai (Jan 2014 – Jul 2014)

- Site Supervision and management in Sinter Plant – 2 Project.
- Design Verification and Technical Comparisons of Electrical Equipments.

Training

✓ **Training on Development of Competency based learning package on S-7 400 PLC**
Tata Steel Limited, Jamshedpur (May 2012 – Jul 2012)

- Learning Platform development for PLC.

✓ **Training on Transmission & Distribution System**
Odisha Power Transmission Corporation Limited (OPTCL), Bhubaneswar (Dec 2011)

- Studying different types of transmission lines and distribution lines. Devices used in switch yards.

- ✓ **Industrial Automation (PLC)**
  Central Tool Room and Training Centre (CTTC), Bhubaneswar (May 2011)
  - A course on Industrial Automation using PLC.

Ph.D. Work

- ✓ **Interlinking Converter Control in Hybrid Microgrid**
  - Identifying the behavior of interlinking converter in hybrid Microgrid and it's similarly with the tie-line connected between two area power system network.
  - Control of parallel operated interlinking converter connected between the AC and DC buses.
  - Controller Hardware-in-loop (CHIL) testing between STM32-F446RE microcontroller and Opal-RT to verify the parallel operated interlinking converter.
  - Improved power balance between AC and DC sub-grids through enhanced hybrid and inverses Q-V droop control with verification through MATLAB simulation.
  - Dual virtual inertia control (DVIC) for the interlinking converter to improve the ROCOF and ROCOV in the hybrid Microgrid and validated by MATLAB simulation.

- ✓ **Centralized Control of Hybrid Microgrid**
  - Modeling and stability of generalized hybrid Microgrid with hierarchical control consisting of primary and secondary control.
  - Primary control includes the droop control whereas the secondary control includes centralized control for frequency and PCC bus voltages restoration.
  - Validation of the control through simulation and verification with CHIL along with use of UART protocol for communication.

- ✓ **Distributed Control of Hybrid Microgrid**
  - Distributed Multi-objective Control of Hybrid Microgrid in Autonomous Mode with verification through MATLAB simulations.
  - Modeling and stability of the generalized hybrid Microgrid with distributed secondary control along with the validation with CHIL and UART protocol for communication.

- ✓ **Frequency Regulation in grid Connected DFIG-Based Wind Turbines**
  - FO-VSG control for grid connected DFIG-Based Wind Turbines to improve ROCOF and frequency nadir. Validations are performed through MATLAB simulation.
  - Enhancing Grid Stability through Adaptive Damping and Inertia Control in DFIG-Based Wind Turbine with Vsync along with validation through MATLAB simulation.

- ✓ **Improving GMPP tracking in Solar PV under partial shading**
  - An effective way of tracking the global maximum power point (GMPP) tracking under partial shading condition and verification through MATLAB simulation.
  - The improvements are performed by identifying the GMPP region in Solar PV characteristics.

Research Publications

- ✓ **Journal**
  - S. Agrawal, B. Tyagi, V. Kumar and P. Sharma, "Digital Controller Design and Implementation for AC and DC Side of 3ph qZSI," in IEEE Transactions on Industry Applications, vol. 60, no. 1, pp. 672-683, Jan.-Feb. 2024.
  - S. Agrawal, B. Tyagi, V. Kumar and P. Sharma, "Operation and Control of Parallel-Operated Interlinking Converter in Hybrid Microgrid," in IEEE Transactions on Industry Applications.
  - S. Agrawal, B. Tyagi, V. Kumar and P. Sharma, "Enhancing Transient Response of Grid Connected DFIG-WT through Adaptive Inertia and Damping Control with Virtual Synchronous," in IEEE Transactions on Industry Applications.

- ✓ **Conferences**
  - S. Agrawal, B. Tyagi, V. Kumar and P. Sharma, "Dual Virtual Inertia Control of Interlinking Converter in Hybrid Microgrid," IEEE PEDES 2024, NIT Surathkal, December 18th to 21st, 2024.
  - S. Agrawal, B. Tyagi, V. Kumar and P. Sharma, "Interlinking Converter Operation with Enhanced Hybrid and Inverse

Q-V Droop," 2024 IEEE PES ISGT Europe, Croatia, October 14^{th} to 17^{th}, 2024.

- S. Agrawal, B. Tyagi, V. Kumar and P. Sharma, "Enhancing Frequency Regulation in DFIG-Based Wind Turbines with FO-VSG," 2024 IEEE Power & Energy Society General Meeting (PESGM), Seattle, Washington, USA, July 21^{st} to 25^{th}, 2024.
- S. Agrawal, B. Tyagi, V. Kumar and P. Sharma, "Enhancing Grid Stability through Adaptive Damping and Inertia Control in DFIG-Based Wind Turbine with Vsync," 2023 IEEE International Conference on Energy Technologies for Future Grids, Wollongong, Australia, December 3-6, 2023.
- S. Agrawal, U. Malik, Y. Tripathy, B. Tyagi, V. Kumar and P. Sharma, "Distributed Multi-objective Control of Hybrid Microgrid in Autonomous Mode," 2023 IEEE Power & Energy Society General Meeting (PESGM), Orlando, FL, USA, 2023, pp. 1-5.
- S. Agrawal, B. Tyagi, V. Kumar and P. Sharma, "Interlinking Converter Operation in Hybrid Microgrid As a Tie-line," 2022 IEEE International Conference on Power Electronics, Drives and Energy Systems (PEDES), Jaipur, India, 2022, pp. 1-6.
- S. Agrawal, B. Tyagi, V. Kumar and P. Sharma, "3ph qZSI Controller Design using PI for DC side and PR for AC side Controller including Droop Characteristic in standalone mode," 2022 IEEE International Conference on Power Electronics, Smart Grid, and Renewable Energy (PESGRE), Trivandrum, India, 2022, pp. 1-6.
- S. Agrawal, B. Tyagi, V. Kumar, P. Agarwal and P. Sharma, "A Simplified and Effective GMPP Tracking Algorithm for Solar Photovoltaic System," 2019 North American Power Symposium (NAPS), Wichita, KS, USA, 2019, pp. 1-6.
- S. Singh, S. Agrawal, and D. N. Vishwakarma, "Online Differential Protection Methodology Based on DWT for Power Transmission System BT - Machine Intelligence and Signal Analysis," 2019, pp. 695–706.

Projects

✓ **Development of Digital Library for DSPs**
Self Project (Aug 2022 – Dec 2022)

- Digital Control Library (DCL): AC Droop, Hybrid Droop, DC Droop, LPF, PR Controller, PI with anti-windup.
- Digital Power Library (DPL): ABC_to_DQ0, DQ0_to_ABC, 3Ph_PLL_SRF, 3Ph_Power, RAMP Generator.
- General Purpose Library (GPL): Update PWM, Update PWM with Limit, Data Frame Extraction from USART.

✓ **JRF in Department of Science and Technology (DST) sponsored project | Funding Rs. 293 Lakh**
SRIC – IIT Roorkee (Oct 2018 – Jan 2022) | Website: https://iitr.ac.in/hmg/

- Hybrid Microgrid Lab-setup at Electrical Engineering Department IIT Roorkee.
- PCB design for gate driver, sensor board and trip circuit.
- Design of converters.
- Installation and testing of electrical components.

✓ **Numerical Protection of Transmission Line Using Spectral Energy Scheme**
M.Tech (Aug 2015 – Jul 2016)
This project involves an analysis of various types of faults in transmission lines and the development of a numerical protection system using a differential spectral energy scheme. The approach includes calculating differential spectral energy in the current signal using Discrete Wavelet Transform (DWT) and employing a sliding data window equivalent to a half cycle of the power system frequency for rapid relaying. The DWT decomposes the signal into different frequency bands for fault detection. The project's performance is tested using MATLAB and SIMULINK under considering different fault conditions.

✓ **Load Flow Solution for Meshed Distribution Networks**
B.Tech (Aug 2012 – Apr 2013)
This study applies power flow analysis to distribution systems, which involve lower-voltage networks connected to various loads. The research introduces an efficient power flow method utilizing current injection and Kirchhoff's laws. This method outperforms traditional approaches like Newton-Raphson and Fast Decoupled Method in terms of convergence. It can handle unbalanced (three-phase) and balanced (single-phase) network representations. The primary objective is to evaluate the Forward-Backward Sweep method's efficiency in solving load flow challenges in meshed distribution networks.

Awards & Academic Achievements

- COMET'25 Poster Presentation 1st Runner-up at IIT Roorkee.
- IEEE PEDES 2024 Som Nath Mahendra Memorial STA Award.
- Hackathon 1st Runner-up in Power System Cybersecurity 2024 hosted by the WRD&M at the IIT, Roorkee, in collaboration with the Centre od Sustainable Energy (CFDE), iHUB DivyaSampark, IIT Roorkee, and THDC India Limited Official.
- International Travel Support (ITS) grant by Anusandhan National Research Foundation (ANRF) formerly known as Science and Engineering Research Board (SERB), Department of Science and Technology, Government of India to attend the IEEE PES ISGT Europe 2024 conference at Croatia .
- Best conference paper submitted to 2023 IEEE PES General Meeting.
- GATE All India Rank (AIR) 2014: 617, GATE All India Rank (AIR) 2013: 1295
- 1st Position in Circuit-O-spice (EED) event at Annual Techno-Management Fest 2011, NIT Rourkela.
- 2nd Position n SETU at Annual Techno-Management Fest 2011, NIT Rourkela.

Position of Responsibility & Extra Curricular

✓ **Cofounder of a Startup – "ATMGARAGE"**
ATMGarage Technologies Pvt. Limited (May 2016 – Sep 2019) | Current Status - Closed
The company was registered as "ATMGARAGE TECHNOLOGIES PRIVATE LIMITED" on 10-05-2016. The company aimed to provide customers with efficient and cost-effective vehicle service. Simultaneously perform R&D in EVs with the generated revenue. The company was also recognized by DIPP with certificate No. DIPP2082. Responsibilities assigned as cofounder: perform R&D, identifying customers' issues and rectifying the concern issues, Account and fund management, website development, and updates.